ALGORITHMS

Week 2

OBJECTIVES

- Understand what an algorithm is
- Understand what algorithms are used for
- Interpret algorithms as flowcharts, pseudocode and written descriptions
- Use and describe the purpose of arithmetic operators
- Understand how to code an algorithm in a high-level language

ALGORITHM

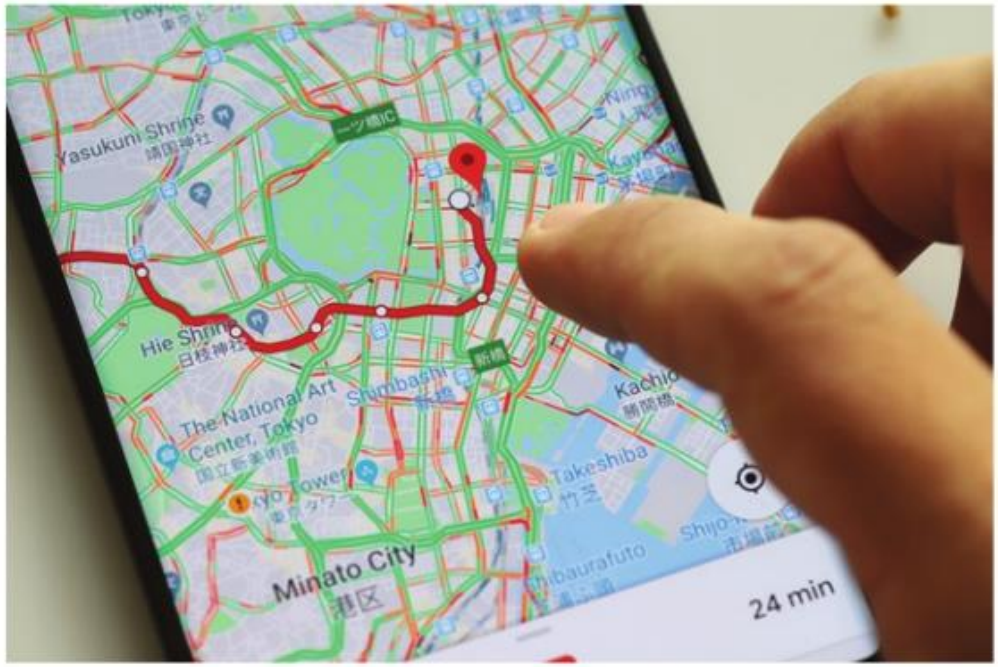
A precise method for solving a problem

EXPLANATION

The route on this interactive map has been calculated using an algorithm.

- It is unambiguous in telling the driver exactly what to do, like 'turn left', 'turn right' or 'go straight'.
- It is a sequence of steps. It can be used again and will always provide the same result.
- It provides a solution to a problem, in this case, how to get from Beijing to Shanghai

A solution to a problem with these characteristics is called an **algorithm**.



SUCCESSFUL ALGORITHMS

There are three points to consider when deciding whether an algorithm is successful or not.

1. **Accuracy** - it must lead to the expected outcome.
2. **Consistency** - it must produce the same result each time it is run.
3. **Efficiency** - it must solve the problem in the shortest possible time, using as few computer resources as possible.

THE RELATIONSHIP BETWEEN ALGORITHMS AND PROGRAMS

Algorithms and programs are closely related, but they are not the same. An algorithm is a detailed design for a solution; a program is when that design is implemented.

EXAMPLE: WRITTEN DESCRIPTIONS

A written description is the simplest way of expressing an algorithm.

ALGORITHM FOR MAKING A CUP OF COFFEE



1. Fill kettle with water.
2. Turn on kettle.
3. Place coffee in cup.
4. Wait for water to boil.
5. Pour water into cup.
6. Add milk.
7. Stir.

ACTIVITY 1

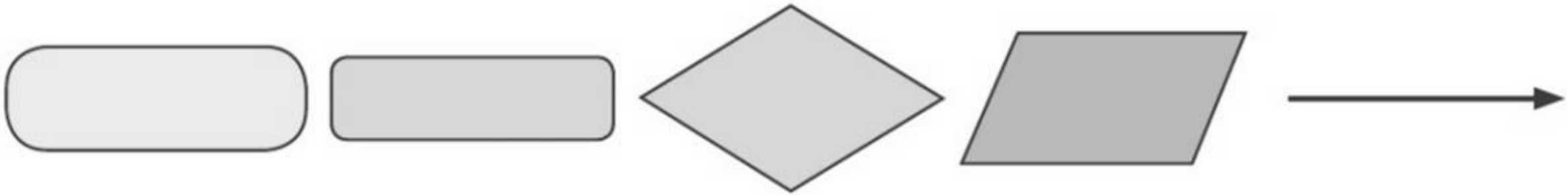
GETTING TO SCHOOL: WRITTEN DESCRIPTION

- Produce a **written description** of an algorithm for getting to school. It should start with leaving home and end with arriving at school. For example, the algorithm could start with 'Walk to bus stop'.

FLOWCHARTS

Shows an algorithm as a diagram. Each step in the algorithm is represented by a symbol. Symbols are linked together with arrows showing the order in which steps are completed

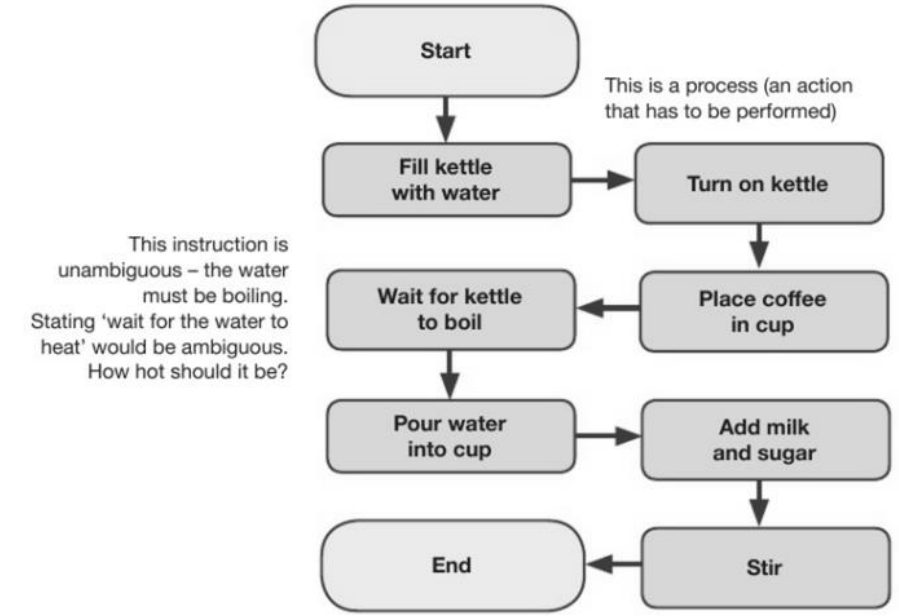
FLOWCHART SYMBOLS



Indicates the start or end of an algorithm	Indicates a process to be carried out	Indicates a decision to be made	Indicates an input or output	Shows the logical flow of the algorithm

EXAMPLE: FLOWCHART

Flowchart of an algorithm to make a cup of coffee



ACTIVITY 2

GETTING TO SCHOOL: FLOWCHART

- Display the 'journey to school' algorithm that you created in Activity 1 as a flowchart.

BONUS: ACTIVITY

BATH FLOWCHART

Display the following as a flowchart. You may need to change the order or add actions.

1. Put in the plug.
2. Fill the bath to the correct level.
3. Check the temperature is OK.

CHECK PAGES (4 – 7)